HIGHWAY INTELLIGENCE · PUNJAB

The Winter Death Trap.

Smog makes drivers outdrive what they can see.

SafarGuard

Javeria Farhan · Hacktoberfest
Hack Day Lahore 2026



Multan–Sukkur Motorway · Rescue 1122 · Dawn, 9 Dec 2025

THE GROUND REALITY

Smog doubles braking distance

+34.8%

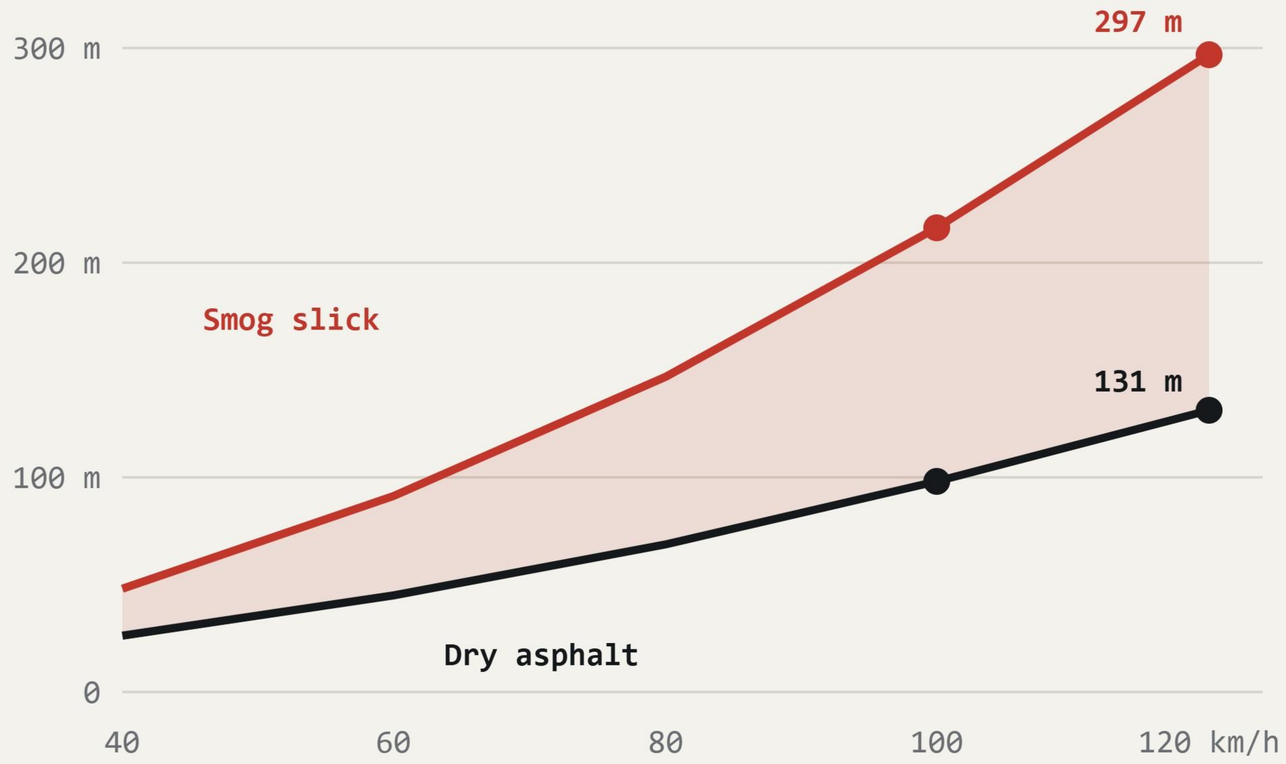
winter highway collisions, Rescue 1122

0.70→0.28

road friction on chemical dew

2.85 s

driver perception delay in smog



SafarGuard · Source: AASHTO standards, Punjab Rescue 1122 benchmarks

THE GAP · 100 KM/H

You see 28 m. You need 216 m.



SafarGuard · Bars drawn to scale

THE SCIENCE

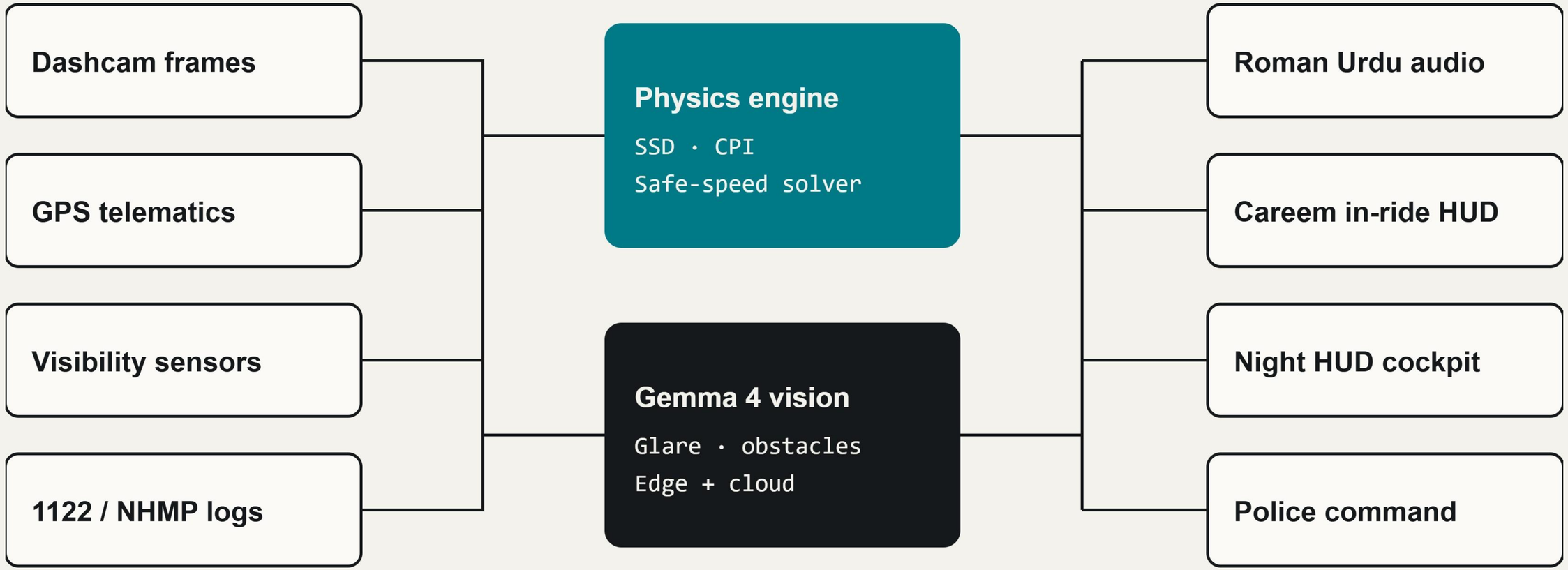
CPI = SSD / Visibility

CPI above 1.0 means a crash is guaranteed. SafarGuard caps speed until it drops below 1.

$$SSD = v \cdot tr + v^2 / (2 \cdot g \cdot f)$$

SafarGuard · AASHTO Stopping Sight Distance

System architecture



SafarGuard · Edge-first · REST · MIT license

LIVE DEMO · ACOUSTIC BEACON

"Khatra shadeed hai! Raftaar foran 35 km/h karein!"

Drivers in fog cannot look at a screen. The cabin speaks.



Zero preventable highway deaths.

Expand
Karachi · M-1 · M-5

Fleets
Daewoo · Faisal Movers · TCS

Contribute
100% MIT open source

`github.com/javeriafarhan/SafarGuard`